

Omar Sadek

Product Engineer

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SUMMARY

Product engineer, shipping production web apps since 2023. I build the interface and the systems under it.

At **Instatus** I own the chat integrations on a status page serving **10M+ visits a month** for Sketch, Harvard, Siemens and Yum! Brands; nights, I build and operate **Wholana** solo: seven deployables, eleven services, one Postgres, on hardware I run myself.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C, Bash

Frontend: React, Next.js, Tailwind, Shadcn UI, HTML/CSS

Backend: Node.js, Flask, Prisma, ZenStack, multi-tenancy, message queues (BullMQ), durable workflows (Temporal), REST APIs, GraphQL

Data: Postgres, Redis, SQLite, vector databases, Scrapy (large-scale crawling)

Cloud & DevOps: AWS (LightSail, S3), DigitalOcean, Hetzner, Cloudflare R2, Linux server admin & SSH, Docker, Docker Swarm, Traefik, CI/CD (Vercel, Railway, Buddy, Dokploy), Git, Sentry

Testing: Unit testing, integration testing, end-to-end testing, Prisma testing patterns

AI / LLM: Prompt engineering (Anthropic guidelines), chain-of-thought design, RAG with domain-knowledge docs, agent skill design, Claude, Gemini, LLM cost optimization, TensorFlow

Tools: Linux, Figma, monorepos

Design: UX/UI, design systems, typography, Figma prototyping

Motion & Interaction: Framer Motion, Web Animations API, CSS Animations; easings, spring vs tween, interruptible animations, clip-path transitions (animations.dev by Emil Kowalski)

EXPERIENCE

Full-Stack Software Engineer, Instatus

Dec 2024 – Present

- Work across the full stack on a product serving **10M+ visits per month** with customers including **Sketch, T2, Harvard University, Siemens, and Yum! Brands**: product design, frontend, backend, database, and AWS cloud infrastructure (LightSail, S3).
- Designed and shipped a **BullMQ message-queue debounce mechanism** for high-frequency backend events, collapsing bursts of hundreds of redundant jobs into a single downstream write.
- Built durable on-call workflows on **Temporal**, hardening the on-call product's reliability and scheduling guarantees.
- Designed a **deltas mechanism** for synchronizing large response payloads against a periodically refreshed base log, cutting incremental update payloads by roughly an order of magnitude.

- **Owned Instatus for Slack end-to-end**, from product strategy and Figma design through implementation: took it from notify-only to running the full incident lifecycle in-channel (declare, post updates, change status, resolve), with auto-created incident channels and templated updates. Now live to all Instatus customers.
- Built the **Instatus Slack AI bot**: deep product-aware assistant for incident response, with prompts engineered against Anthropic's guidelines for chain-of-thought reasoning, RAG-style grounding in internal domain-knowledge docs, and aggressive cost optimization on the model loop. Evaluated model output quality and edge-case failure modes before shipping.
- Reworked core systems including uptime percentage calculations and incident handling to make the product more reliable.
- Shipped improvements to on-call scheduling, monitoring, incident channel creation, and templating across the product.
- Strengthened on-call scheduling test coverage and surfaced multiple previously-undiscovered bugs in the scheduling system.
- Built features against GraphQL APIs and Redis-backed caching layers; integrated AWS S3 for media and asset storage.
- Established Prisma-based unit, integration, and end-to-end testing patterns and contributed to backend reliability and performance optimization.

Founding Engineer, Loom Cairo (later Univyr)

Part of **AUC Venture Lab (V-Lab)** Cycle 25 startup accelerator.

Jun 2023 – Oct 2025

- Architected and shipped, solo, a fashion marketplace and SaaS storefront builder for the Egyptian market: Next.js + TypeScript frontend, Python (Scrapy) ingestion pipelines, Postgres via Prisma, Redis caching, and payment gateway integration.
- **Built the entire commerce backend from scratch, with no Shopify and no off-the-shelf e-commerce OS**, reliably serving orders, addresses, user data, and transactional email for a platform that peaked at **40,000+ unique visitors**.
- Designed the schema for a full **e-commerce OS** inspired by **Medusa.js** (not using) and adapted to local constraints, with **multi-tenancy** and **ZenStack-based authorization** across two user models: marketplace customers, and vendor users tied to organizations inside the storefront-builder OS.
- Built a high-throughput **Scrapy crawler** ingesting **~1M items / ~4M variants across 43,000 websites in ~20 minutes per run** (Univyr, 2025); went deep on Scrapy internals (scheduler, spider, yielding pipelines into the DB, heavy concurrency, proxy rotation) for robust crawls at million-item scale.
- Implemented vector-embedding search; built ML data labeling and image classification with TensorFlow to enrich the catalog.
- Architected the platform: monorepo, Prisma schema, session-based auth, SOLID-based scraper classes with custom context managers for DB locks, exception handling, and logging.
- Provisioned and operated the infrastructure solo: configured **AWS LightSail** for the application and pushed the Scrapy crawler to a **DigitalOcean droplet**, managing both via SSH for deployments, service starts, and ongoing ops.
- Designed the product UI, brand, and landing page from scratch in Figma.

- Grew to **70+ brand partnerships** and **40,000+ unique visitors** at peak, off **4,700 MAU at 76% returning** when pitching V-Lab in June 2025; accepted into **AUC Venture Lab (V-Lab)** startup accelerator. Shut down without raising: the aggregator sent brands free traffic and never captured margin, which is the lesson I built Wholana's billing around first.

Software Engineer (Contract), UNITAR (UN Agency)

Sep 2024 – Nov 2024

- Built the frontend of a React + TypeScript proof-of-concept web app for managing activities across regions, scales, and stakeholders.
- Led UI design in Figma, optimizing layout for a consistent cross-device experience.
- Improved project organization by separating concerns: routing, initialization, and modular components.
- Collaborated in a cross-functional team with agile workflows, sprint planning, Git feature branching, and code review.

Software Engineering Instructor, ITI

Jul 2023 – Aug 2023

- Taught core programming concepts to 30+ high school students: loops, conditionals, functions, program flow.
- Guided students through abstractions, debugging, and best practices.
- Received positive feedback for clear explanations and student "aha" moments.

Summer Academy Intern, Dell Technologies · Aug–Sep 2022

- Docker, RedHat Linux, and Data Center Architectures coursework; team capstone across AI, DevOps, Testing, and 5G.

Access Transmission Engineer Intern, Orange Egypt · Jul–Aug 2022

- Freed ~15% of bit rate on cell sites by phasing out legacy PDH E1 in favor of VLAN-based networking (NEC, Ericsson NMS).

PROJECTS

Wholana · Founder & Engineer · 20 paying users, \$100 MRR

wholana.com · changelog: wholana.com/changelog

A research and ideation platform for short-form video creators: it surfaces high-performing TikToks, decodes their structure (hook, beat sheet, logline, stop-scroll, stakes-value), and helps creators generate new ideas grounded in real data. **14 substantive releases shipped solo between March and July 2026**, on a public changelog.

- Next.js + TypeScript + Prisma + Postgres with pgvector; **multi-tenant workspaces** with a workspace switcher and member invitations; **seat-based Team billing** (monthly and yearly) through Polar; Better Auth with Google OAuth and One-Tap.
- Built the **Wholana Decoder**: a nightly ingestion and enrichment pipeline on **Temporal** durable workflows (scrape, subtitles, LLM craft pass via OpenRouter), emitting structured beat sheets, a closed craft taxonomy, and outlier scoring. **Wrote and self-hosted a TikTok scraper that replaced the paid third-party one**, behind an adapter seam so no downstream module names a vendor.

- Shipped a **Wholana MCP server** (stdio and streamable HTTP) that connects a workspace to ChatGPT and other MCP clients to search the corpus and read back why a video worked; **local-first reads on Rocicorp Zero** with optimistic writes, and **live multiplayer Scripts** on Yjs + Hocuspocus with in-memory presence.
- **Operates the production infrastructure solo**: two Hetzner boxes running Docker Swarm via Dokploy behind Traefik and Cloudflare, R2-backed media on a dedicated CDN with Imaginary, per-PR preview environments, Sentry observability, nightly backups to R2, and cron-staleness alerting.
- Designed and shipped the entire surface area solo: Explore with saved lenses and a filter builder, swipe file, Scripts, creator profiles, dashboards, Billing, admin Control Room, plus the marketing site and in-app brand in an editorial / serif-led style.

Argonaut · Freelance Engineer, Designer & Brand Strategist

Engineering & trading firm · HVAC, MEP, fire safety

- Owned product, brand, and engineering end-to-end. The CRM is now in **daily production use by a 10+ person team running 8-hour shifts** to operate the business.
- Researched the MEP industry deeply to inform brand positioning and stakeholder messaging; defined company values (**reliability, sustainability, innovation**) and built the marketing site in a **corporate-minimalist** design system that surfaces what Argonaut does, its client roster, and its supplier credibility.
- Marketing site **ranks at the top of search results** and **surfaces in AI-generated responses**; client confirms the site has converted real new customers.
- Translated domain research into the CRM's data model: extensible schemas with branching fields that consolidated fragmented spreadsheet workflows into a single robust database.
- Shipped the CRM: quote/RFQ workflows handling multi-million-dollar quote pipelines, registrations, supplier and consultant tracking, dashboards for quotation value, and won/lost/pending metrics.

Argotemp · Freelance Engineer & Designer

Equipment rental & maintenance operations platform

- Built an end-to-end rental and maintenance operations system: rental unit availability, maintenance state machines, monthly invoicing, and job extensions and renewals.

Little Lads · Web Design

Boys' fashion brand

- Revitalized brand identity and storefront to improve brand equity, engagement, and conversion.

EDUCATION

Misr International University, 2018–2023

BSc. Electronics & Communications Engineering

Coursework: Microprocessor-Based Systems, Logic Circuits Design, Programming with C++

Egypt British International School, 2015–2018

IGCSE, British Secondary School Certificate

LANGUAGES

- **English**: C2 (fluent / bilingual)
- **Arabic**: Native

- **French:** B1 (intermediate)

EXTRACURRICULARS

- **Toastmasters Cairo:** 3× winner, Best Table Topics Speaker